

2. Maximum Association Rules and Frequent Itemsets

The PDF asks the theoretical questions about the market-basket transactions.

Steps

1. Write the given transactions in a table.
2. List all possible non-empty item combinations.
3. Generate possible association rules.
4. Calculate the maximum number of rules.
5. Find the maximum possible frequent-itemset size.
6. Write the final answers.

Program code: .arff file

```
@relation market Basket

@attribute M {0,1}
@attribute O {0,1}
@attribute N {0,1}
@attribute K {0,1}
@attribute E {0,1}
@attribute Y {0,1}
@attribute D {0,1}
@attribute A {0,1}
@attribute U {0,1}
@attribute C {0,1}
@attribute I {0,1}

@data
1,1,1,1,1,1,0,0,0,0,0,0
0,1,1,1,1,1,1,0,0,0,0,0
1,0,0,1,1,0,0,1,0,0,0,0
1,0,0,1,0,1,0,0,1,1,0,0
0,1,0,1,1,0,0,0,0,1,1,1
```

OUTPUT :

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Associator: Choose **FPGrowth** -P 2 -I -1 -N 10 -T 0 -C 0.8 -D 0.05 -U 1.0 -M 0.5

Start Stop

Result list (right-click for...)

- 08:43:40 - Apriori
- 09:02:01 - FPGrowth

Associator output

```
=== Run information ===

Scheme: weka.associations.FPGrowth -P 2 -I -1 -N 10 -T 0 -C 0.8 -D 0.05 -U 1.0 -M 0.5
Relation: market_Basket
Instances: 5
Attributes: 11
M
O
N
K
E
Y
D
A
U
C
I

=== Associator model (full training set) ===

FPGrowth found 9 rules (displaying top 9)

1. [E=1]: 4 ==> [K=1]: 4 <conf: (1)> lift: (1) lev: (0) conv: (0)
2. [Y=1]: 3 ==> [E=1]: 3 <conf: (1)> lift: (1) lev: (0) conv: (0)
3. [O=1]: 3 ==> [E=1]: 3 <conf: (1)> lift: (1) lev: (0) conv: (0)
4. [M=1]: 3 ==> [E=1]: 3 <conf: (1)> lift: (1) lev: (0) conv: (0)
5. [O=1]: 3 ==> [E=1]: 3 <conf: (1)> lift: (1.25) lev: (0.12) conv: (0.6)
6. [O=1]: 3 ==> [E=1], [M=1]: 3 <conf: (1)> lift: (1.25) lev: (0.12) conv: (0.6)
7. [E=1, O=1]: 3 ==> [E=1]: 3 <conf: (1)> lift: (1.25) lev: (0.12) conv: (0.6)
8. [E=1, O=1]: 3 ==> [E=1]: 3 <conf: (1)> lift: (1) lev: (0) conv: (0)
9. [E=1]: 5 ==> [E=1]: 4 <conf: (0.8)> lift: (1) lev: (0) conv: (0.5)
```

Status OK Log x0